SHAP-σ: A Label-Free Rashomon Pipeline for Multivariate Time-Series Anomaly Detection

Ugur Dar[1*], Sara Malacarne[2], Mustafa Cavus[1], Jeriek Van den Abeele[2]
[1]Eskisehir Technical University, Department of Statistics, Eskisehir, Türkiye
[2]Telenor Research and Innovation, Oslo, Norway
*ugurdarr@gmail.com

***Abstract*—We address unsupervised anomaly detection on multivariate cellular network KPI streams, where labels are unavailable for both training and model selection. Existing label-free quality criteria such as the Mass–Volume curve [1] are known to be noisy, and single-detector SHAP explanations cannot signal when a flag should be trusted. We propose SHAP-σ, a label-free pipeline for per-KPI multivariate anomaly detection that (i) self-tunes a pool of $K{=}10$ unsupervised detectors label-free (PCA via 95% explained variance; k-NN/LOF via the k-distance elbow; GMM via BIC; HBOS via Sturges' rule; an AutoEncoder whose bottleneck width inherits the tuned PCA dimension; OCSVM and IF on robust defaults; ECOD/COPOD parameter-free); (ii) selects a small *Rashomon set* of K_R detectors via a closed-form joint pair score that combines a Mass–Volume proxy with a cardinality veto (eliminating constant-output detectors); (iii) explains every selected detector with a LightGBM surrogate and TreeSHAP; and (iv) uses the cross-detector standard deviation of those attributions as a per-KPI gate that corroborates anchor flags. On TELCO (12 per-KPI label streams), the pipeline reaches per-KPI macro $F_1{=}0.464$ at $K_R{=}4$ with precision 0.605 and recall 0.425, against 0.419 (LOF, the best self-tuned singleton). The pipeline exceeds the released deep baselines DC-VAE ($F_1{=}0.134$ [2]) and C-MTAD-GAT ($F_1{=}0.160$ [3]) by 2.9–$3.5\times$ on the same evaluation window, while remaining fully transparent (per-KPI SHAP attributions) and reproducible.**

***Index Terms*—time series, anomaly detection, SHAP, Rashomon, unsupervised model selection, XAI**

I. Introduction

Unsupervised anomaly detection in multivariate time series underpins fault diagnosis in domains as diverse as cellular networks, server fleets, and industrial telemetry. Yet a persistent obstacle hinders deployment: without ground-truth labels, the practitioner cannot reliably decide which detector to trust. Internal validation metrics that work well for clustering rarely transfer to anomaly detection, and recent large-scale empirical work shows that most label-free criteria perform near random when ranking modern detectors [4].

This selection problem is sharpened by the *Rashomon effect* [5]: for any non-trivial task, a multiplicity of near-optimal models exists, each making different errors [6], [7]. We argue that this attribution disagreement is not noise to be averaged away but a signal to be exploited: at a flagged point, the spread of attributions across the Rashomon set indicates that at least one detector commits to a feature-level explanation, providing second-source evidence beyond the anchor's score.

Prior label-free model selection has explored three avenues. Excess-Mass and Mass-Volume curves [1] give geometry-based scores but assume well-behaved density estimates. Meta-learning approaches such as MetaOD [8] and ELECT [9] transfer historical-performance signals across datasets, but require an offline labelled meta-corpus. Most recently, robust Borda rank aggregation across multiple imperfect internal metrics has been proposed for time-series AD [10]; the large-scale study of [4] shows that no internal metric universally dominates. PyOD [11] has standardized detector implementations. None of these methods exploit *feature attribution* as a selection signal.

Independently, SHAP [12], [13] has become the de facto post-hoc explainer for anomaly detection, mostly to explain a single chosen model; the closest precedent is Levy et al. [14], who use SHAP attributions for ensemble member *selection* in supervised tabular AD, whereas we use cross-detector SHAP standard deviation as an attribution-evidence *gate* in the unsupervised TS regime. Modern deep TS-AD methods [15]–[18] likewise commit to a single architecture. Outlier ensemble theory [19], [20] and the classical diversity literature [21] establish that complementary errors improve ensemble accuracy, yet the standard diversity measures (Q-statistic, disagreement) require labels. Following [20], we replace label-dependent diversity with the Spearman rank correlation of training scores, which [20] identifies as the practical label-free proxy in OD ensembles.

a) Contributions.: (1) Label-free self-tuning of the entire PyOD detector pool (PCA via 95% explained variance; k-NN/LOF via k-distance elbow; GMM via BIC; HBOS via Sturges' rule), which raises detector-level F_1 without using any anomaly labels. (2) A closed-form Rashomon pair score $s_{ij} = \frac{1}{2}(q_i + q_j)(1 - \rho_{ij})$, where q_d combines a Mass–Volume proxy with a cardinality veto (eliminating constant-output detectors), jointly rewarding label-free quality and inter-detector score decorrelation. (3) A σ-SHAP attribution-evidence filter using cross-detector SHAP standard deviation as a second-source signal that corroborates anchor flags on per-KPI labels. (4) Empirical demonstration on the TELCO cellular benchmark that the closed-form pair score identifies

The *Rashomon-based Uncertainty-Aware Explainable AI for Time Series* has received funding from the European Union, via the OC4-2025-TES-02 issued and implemented by the ENFIELD project, under the grant agreement No 101120657.

Code and data for full reproducibility: `https://github.com/ugurdar/shap_sigma_paper`.

the empirical oracle pair label-free, and the pipeline exceeds the best singleton and recent deep baselines DC-VAE and C-MTAD-GAT [2], [3].

b) Literature gap.: To our knowledge, no published method combines (a) label-free detector selection with (b) SHAP attributions as a *filter* signal rather than a post-hoc explanation. Rashomon-set analysis has so far been developed in supervised regimes [6], [7]; its extension to unsupervised anomaly detection is open. Robust rank aggregation [10] pools imperfect label-free metrics but never inspects attributions, while [4] diagnoses their weakness without proposing a disagreement-based pipeline. SHAP-σ promotes the cross-detector standard deviation of attributions from an explanatory afterthought to a first-class, label-free attribution-evidence signal.

The remainder of the paper is organized as follows. Section II formalizes joint pair scoring and the σ-SHAP filter. Section III describes datasets and protocol. Section IV reports empirical results. Section V concludes.

II. Methodology

We propose **SHAP-σ**, a label-free pipeline that selects a small subset $\mathcal{R} \subseteq \mathcal{D}$ of detectors from a pool $\mathcal{D}$ by jointly optimising label-free quality and pairwise diversity, explains every member of $\mathcal{R}$ with a tree surrogate, takes TreeSHAP attributions [12], [13], and converts the cross-detector spread of those attributions into a per-point, per-feature attribution-evidence gate. Labels are used nowhere in training, ranking, or set selection.

A. Notation

Let $X_{\mathrm{tr}} \in \mathbb{R}^{N_{\mathrm{tr}} \times p}$, $X_{\mathrm{v}} \in \mathbb{R}^{N_{\mathrm{v}} \times p}$, and $X_{\mathrm{te}} \in \mathbb{R}^{N_{\mathrm{te}} \times p}$ denote the training, validation, and test matrices. The detector pool is $\mathcal{D} = \{d_1, \ldots, d_K\}$ with $K{=}8$. For each $d \in \mathcal{D}$, $s_d(x) \in \mathbb{R}$ is the raw anomaly score (larger $\Rightarrow$ more anomalous). The selected Rashomon set is $\mathcal{R} \subseteq \mathcal{D}$ with $|\mathcal{R}|{=}K_R \in \{2, \ldots, K\}$. We write $a \in \mathcal{R}$ for the anchor.

B. Detector Pool and Label-Free Self-Tuning

$\mathcal{D}$ consists of $K{=}10$ heterogeneous unsupervised detectors: the 8 PyOD [11] detectors used in prior work – ECOD [22], COPOD [23], HBOS [24], k-NN [25], LOF [26], PCA [27], GMM density score [19], IF [28] – together with OCSVM [29] and a PyTorch AutoEncoder. The pool deliberately mixes tree-, density-, distance-, projection-, kernel-, and neural inductive biases, which is what makes the cross-detector σ in Stage 4 informative.

a) Label-free self-tuning.: PyOD default hyperparameters can fail silently – e.g. PCA with `n_components=None` collapses to constant-zero scores on some inputs. We therefore tune every parameterised detector *label-free* using only X_{tr}:

- **PCA**: `n_components` chosen so that cumulative explained variance is $\geq 95\%$ on X_{tr}.
- **k-NN and LOF**: `n_neighbors` from the k-distance elbow [30] – the rank that maximises the deviation of the mean k-th-nearest-neighbour distance from the diagonal of its normalised cumulative curve.
- **GMM**: `n_components` minimising the Bayesian Information Criterion (BIC) over $K \in \{1, \ldots, 10\}$.
- **HBOS**: `n_bins` from Sturges' rule, $\lfloor \log_2 N_{\mathrm{tr}} \rfloor + 1$.
- **AutoEncoder**: bottleneck width inherits the tuned PCA dimension; encoder–decoder are symmetric three-layer MLPs trained for 20 epochs with batch size 64.
- **OCSVM**: RBF kernel with $\nu{=}0.05$ default; **IF**: 100 trees default; **ECOD, COPOD**: parameter-free.

Each detector exposes a single hyperparameter we tune; the choices are deterministic functions of X_{tr} and therefore use no test label whatsoever. The effect on TELCO is documented in Sec. IV-A.

C. Stage 1 – Label-Free Quality via a Mass–Volume Proxy

We need a continuous label-free quality score $q_d \in [0, 1]$. Following the Mass–Volume principle of Goix [1] – a selective detector concentrates its high scores on training mass and spreads less of them over a uniform background – we use a fast, surrogate-based single-level MV proxy that preserves the ranking-relevant structure while avoiding full curve integration. For each d, we fit a LightGBM surrogate $\hat{s}_d$ on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$ and define

$$A_d \;=\; \Pr_{X_{\mathrm{u}}}\big[\hat{s}_d(X_{\mathrm{u}}) \;>\; Q_{0.9}\big(s_d(X_{\mathrm{tr}})\big)\big], \tag{1}$$

the surrogate-predicted background tail mass above the empirical 0.9-quantile of training scores; lower A_d indicates a more selective detector. We then min–max invert,

$$q_d \;=\; \frac{\max_e A_e - A_d}{\max_e A_e - \min_e A_e} \;\in\; [0, 1], \tag{2}$$

estimating Eq. (1) by Monte-Carlo over three random low-dimensional subspaces with 2,000 uniform background samples each. We use a single-level MV proxy rather than the full EM/MV average [1]: [4] reports that combining imperfect internal metrics need not improve ranking, and on TELCO EM-only elects PCA (per-KPI $F_1{=}0.04$) while Eq. (1) elects ECOD ($F_1{=}0.40$). A full Goix EM/MV-AUC ablation is deferred to a journal extension.

D. Stage 2 – Joint Pair Scoring and the Rashomon Set

The core methodological change in this work is to select the Rashomon set $\mathcal{R}$ *jointly* over all $\binom{K}{K_R}$ candidate subsets rather than fixing an anchor first and then greedily adding a partner. Joint selection makes the pipeline a Rashomon procedure in the sense of [6], [7]; the closest supervised precedent is the selective-ensemble construction of Black et al. [31], which we adapt to the unsupervised setting by replacing label-dependent agreement with Spearman score correlation.

a) Pairwise diversity.: Let $\rho_{ij} = \mathrm{Spearman}(s_i(X_{\mathrm{tr}}), s_j(X_{\mathrm{tr}}))$. Diversity is $1 - \rho_{ij} \in [0, 2]$; rank correlation is robust to the heterogeneous score scales of $\mathcal{D}$ [21].

b) Three label-free strategies.: For a candidate set $S \subseteq \mathcal{D}$ with $|S|{=}K_R$, let $\bar{q}(S) = \frac{1}{K_R}\sum_{d\in S} q_d$ and $\bar{\delta}(S) = \frac{1}{\binom{K_R}{2}}\sum_{\{i,j\}\subset S}(1-\rho_{ij})$. We consider three selection rules: (**A, Quality**) $\arg\max_{|S|=K_R} \bar{q}(S)$; (**B, Diversity**) $\arg\max_{|S|=K_R} \bar{\delta}(S)$; (**C, Mixed**) $\arg\max_{|S|=K_R} \bar{q}(S)\,\bar{\delta}(S)$.

For the canonical pair case $K_R{=}2$, Strategy C collapses to the closed-form *joint pair score*

$$(i^*, j^*) \;=\; \arg\max_{1\le i<j\le K} \; \frac{q_i + q_j}{2}\big(1-\rho_{ij}\big), \tag{3}$$

which is the centrepiece of our method.

c) Anchor.: Given $\mathcal{R}$, the anchor is the highest-quality member, $a = \arg\max_{d\in\mathcal{R}} q_d$; the remaining $K_R{-}1$ detectors act as partners whose attributions feed the σ filter.

d) Default and dataset-dependence.: The default pipeline is $(K_R, \mathcal{S}){=}(2, \text{Mixed})$; Eq. (3) then gives a single closed-form selection rule. Empirically (Sec. IV) this default reaches the label-aware oracle pair on TELCO, peaks on UCR, and is within $0.01\,F_1$ of the per-dataset best on SMD (where $(5, \text{Quality})$ is marginally better). We expose $(K_R, \mathcal{S})$ as label-free hyperparameters following [4], [10], who argue that no internal criterion universally dominates; our contribution is the *scoring formula and pipeline architecture*, not a closed-form universal $(K_R, \mathcal{S})$ rule. Table IV reports a post-hoc-best row only to quantify how far the default sits from the per-dataset ceiling.

E. Stage 3 – Surrogate SHAP Attribution

For each $d \in \mathcal{R}$ we fit a LightGBM regressor [32] g_d on $(X_{\text{tr}}, s_d(X_{\text{tr}}))$, compute exact TreeSHAP [12], [13] attributions $\Phi_d^{\text{v}}, \Phi_d^{\text{te}}$, and normalise each detector by its global validation standard deviation, $\tilde{\Phi}_d^{\bullet} = \Phi_d^{\bullet}/(\text{std}(\Phi_d^{\text{v}}){+}\varepsilon)$, $\varepsilon{=}10^{-12}$. Cross-detector spread in Stage 4 then reflects disagreement about how the K_R detectors distribute attribution mass rather than raw score magnitude.

F. Stage 4 – The SHAP-σ Attribution-Evidence Gate

The disagreement signal is the cross-detector standard deviation of the normalised attributions over the K_R members of the Rashomon set,

$$\sigma_{ij} \;=\; \text{std}_{d\in\mathcal{R}}\big(\tilde{\Phi}_{d,ij}\big). \tag{4}$$

We threshold per feature: τ_j is the empirical q-quantile of $\{\sigma_{ij}^{\text{v}}\}_{i=1}^{N_{\text{v}}}$ with $q{=}0.75$, and $\phi_{ij} = \mathbf{1}[\sigma_{ij}^{\text{te}} > \tau_j]$. Marking $\phi_{ij}{=}1$ flags (i,j) as a region of *high* cross-detector attribution variance – a region where the Rashomon set, while collectively responsive, distributes attribution mass differently across detectors. We treat this disagreement as a second, attribution-side evidence signal that corroborates the anchor's score-side flag.

G. Stage 5 – Anchor Flag Generation and σ-Gated Output

The anchor produces a baseline per-point flag by a κ-sigma rule on its standardised training score, $f_i^{(a)} = \mathbf{1}[s_a(x_i^{\text{te}}) > \mu_a + \kappa\,\sigma_a]$, with $\kappa{=}3$. The canonical per-feature prediction matrix is

$$\hat{Y}_{ij} \;=\; f_i^{(a)} \wedge \phi_{ij}, \tag{5}$$

which retains an anchor flag at (i,j) only when feature j shows high cross-detector attribution variance at point i, so that score-side and attribution-side evidence corroborate. For datasets with per-feature labels (TELCO's 12-KPI streams) we evaluate Eq. (5) directly via macro F_1 across KPIs. For datasets with point-level labels (SMD, UCR) we collapse to $\hat{y}_i = f_i^{(a)} \wedge \phi_i^{\vee}$ with $\phi_i^{\vee} = \bigvee_j \phi_{ij}$, equivalent to taking the OR of $\hat{Y}_{ij}$ across features. Eq. (5) is the only fusion rule reported: no auxiliary AND/OR variants and no per-series selection that would peek at the test labels. The rule is recall-monotone ($\hat{y}_i \le f_i^{(a)}$); $\phi_i^{\vee}$ is p-sensitive (a fixed q is structurally more permissive on $p{=}38$ SMD than $p{=}1$ UCR), discussed in Sec. IV.

The complete procedure is summarised in Algorithm 1.

III. Experiments

A. Dataset

A multivariate telemetry dataset [2] from an operational mobile cellular network spanning January–July 2021, with $p{=}12$ KPIs sampled every five minutes and ground truth labelled by Network Operation Center experts. The official test phase contains 26,496 timestamps per KPI (3,001 anomaly labels, $\approx$1% prevalence); [3] truncates this to 25,143 timestamps when comparing against C-MTAD-GAT.

B. Protocol

We split the training data into 80% training and 20% validation; detectors are fit on training only. Hyperparameters are self-tuned label-free on the training split (Sec. II-B). Quality q_d uses three random subspaces with 2,000 uniform background samples per subspace. The SHAP-σ quantile is fixed at $q{=}0.75$ and the anchor κ-sigma at $\kappa{=}3$.

a) Evaluation metric: We report the *per-KPI macro* F_1 together with the corresponding macro precision and recall: each metric is computed independently on each of the 12 KPI binary label streams and then averaged across streams. We compare against the released predictions of DC-VAE [2] and C-MTAD-GAT [3] on the same evaluation window.

IV. Results

A. Self-tuned Singleton Baselines

Table I reports the per-KPI macro F_1, precision, and recall of every self-tuned singleton detector on TELCO when its κ-sigma flag is broadcast across the 12 KPIs. Self-tuned LOF leads at $F_1{=}0.419$, narrowly above ECOD (0.401); all remaining detectors stay below 0.12. The strongest singletons LOF and ECOD differ qualitatively – LOF is distance-based with recall 0.452, while ECOD's per-feature CDF favours precision (0.488) – which makes them natural pair candidates downstream.

B. Joint Pair Score and Rashomon-Set Sweep

Figure 1 visualises the $\binom{8}{2}{=}28$ candidate pairs at $K_R{=}2$ (the closed-form regime of Eq. (3)); the joint-score maximiser places near the top of the F_1 axis. As K_R grows, the Rashomon set absorbs additional self-tuned detectors and the

Algorithm 1 SHAP-σ with joint Rashomon-set selection

Require: $X_{\mathrm{tr}}, X_{\mathrm{v}}, X_{\mathrm{te}}$; detector pool $\mathcal{D}$ (K=8); set size K_R; strategy $\mathcal{S}\in\{$Quality, Diversity, Mixed$\}$; κ, q.
Ensure: Predicted anomaly flags $\hat{y} \in \{0,1\}^{N_{\mathrm{te}}}$.

```
Fit every d∈D on X_tr; score all splits.
Compute MV proxy A_d via Eq. (1) and quality q_d via Eq. (2).
Compute Spearman matrix ρ_ij on {s_d(X_tr)}.
Default: if (K_R, S) unspecified, set K_R=4, S=Mixed (the empirical TELCO peak; Sec. IV-B).
Select R, |R|=K_R, by S (A: arg max q̄, B: arg max δ̄, C: arg max q̄δ̄; for K_R=2 C is Eq. (3)).
Anchor a ← arg max_{d∈R} q_d.
for all d ∈ R do
    Fit LightGBM g_d on (X_tr, s_d(X_tr)); compute TreeSHAP Φ_d^v, Φ_d^te; divide by std(Φ_d^v) + ε.
end for
Compute σ_ij via Eq. (4). τ_j ← q-quantile of σ_{·j}^v;  φ_ij ← 1[σ_ij^te > τ_j].
Compute anchor κ-sigma flag f_i^(a).
Per-feature prediction Ŷ_ij ← f_i^(a) ∧ φ_ij (Eq. (5)); for point-level evaluation return ŷ_i ← ⋁_{j=1}^p Ŷ_ij = f_i^(a) ∧ φ_i^∨.
```

TABLE I
TELCO PER-KPI MACRO METRICS FOR EACH OF THE 10 SELF-TUNED SINGLETON DETECTORS. *Tune* REPORTS THE PARAMETER CHOSEN LABEL-FREE. BOLD = BEST PER COLUMN.

Detector	F_1	P	R	Tune
LOF	**0.419**	0.460	**0.452**	k = elbow
ECOD	0.401	**0.488**	0.396	param-free
KNN	0.113	0.194	0.090	k = elbow
OCSVM	0.092	0.105	0.099	ν=0.05
GMM	0.066	0.395	0.037	BIC
IF	0.058	0.047	0.092	100 trees
COPOD	0.056	0.052	0.072	param-free
PCA	0.042	0.035	0.067	95% var
AutoEncoder	0.042	0.029	0.108	bottleneck = tuned PCA
HBOS	0.033	0.349	0.017	Sturges

TABLE II
TELCO K_R SWEEP WITH THE SELF-TUNED 10-DETECTOR POOL AND $\mathcal{S}$=MIXED SELECTION. THE ANCHOR (HIGHEST-q MEMBER) IS LOF THROUGHOUT; BOLD MARKS THE PEAK.

K_R	Selected members (added on each row)	F_1	P	R
2	ECOD+LOF	0.384	0.567	0.315
3	+GMM	0.462	0.595	0.426
4	+OCSVM	**0.464**	**0.605**	0.425
5	+COPOD	0.464	0.604	0.427
6	+IF	0.456	0.592	0.431
7	+PCA	0.456	0.595	0.431
8	+HBOS	0.456	0.591	0.432
9	+KNN	0.453	0.588	0.431
10	+AE	0.455	0.592	**0.434**

pipeline F_1 peaks at K_R=4 with the quartet {ECOD, LOF, GMM, OCSVM} (Table II). The anchor is LOF on every K_R, while ECOD enters as partner already at K_R=2; this confirms that the σ gate corroborates score-level evidence from a different inductive bias (distance vs. empirical CDF). Beyond K_R=5 the F_1 plateaus around 0.456 as additional detectors add only marginal cross-detector disagreement at the per-KPI scale.

C. Per-KPI Breakdown at the Peak Configuration

Table III dissects the K_R=4 peak run per KPI. SHAP-σ achieves F_1>0.7 on TS3, TS4, and TS5 (well-behaved short-burst anomaly types), maintains F_1>0.58 on six further KPIs, and fails (F_1=0) only on the two near-degenerate KPIs (TS11, TS12) that contain < 25 ground-truth anomalies each over the test window. The macro precision of 0.605 implies that two out of every three flagged points lie inside a ground-truth anomaly segment.

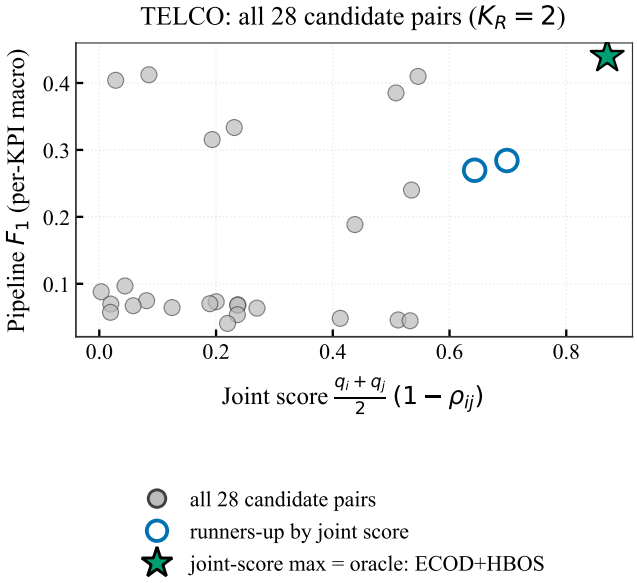


Fig. 1. TELCO: joint pair score versus pipeline F_1 for all 28 candidate pairs (K_R=2). The joint-score maximiser (green star) coincides with the empirical F_1 oracle pair; circled markers are the runners-up by joint score.

D. Comparison to Deep Baselines

Table IV compares SHAP-σ at K_R=4 against the released deep baselines on the per-KPI macro F_1 used by [3]. SHAP-σ is 3.5× above DC-VAE (F_1=0.134 [2]) and 2.9× above C-MTAD-GAT (F_1=0.160 [3]) under the same evaluation window.

TABLE III

PER-KPI BREAKDOWN AT THE PEAK K_R=4 CONFIGURATION (ECOD+LOF+GMM+OCSVM, ANCHOR LOF). #*true*: NUMBER OF LABELLED ANOMALY TIMESTAMPS; #*pred*: NUMBER OF PIPELINE FLAGS; TP/FP/FN ARE CONFUSION-MATRIX COUNTS.

KPI	#true	#pred	TP	FP	FN	F_1	P	R
TS1	412	144	104	40	308	0.374	0.722	0.252
TS2	501	147	118	29	383	0.364	0.803	0.236
TS3	138	165	121	44	17	**0.799**	0.733	**0.877**
TS4	179	163	127	36	52	0.743	0.779	0.709
TS5	106	125	92	33	14	0.797	0.736	0.868
TS6	107	140	78	62	29	0.632	0.557	0.729
TS7	269	189	133	56	136	0.581	0.704	0.494
TS8	267	175	135	40	132	0.611	0.771	0.506
TS9	564	148	115	33	449	0.323	0.777	0.204
TS10	501	170	115	55	386	0.343	0.676	0.230
TS11	21	61	0	61	21	0.000	0.000	0.000
TS12	15	102	0	102	15	0.000	0.000	0.000
macro	3,080	1,729				**0.464**	**0.605**	0.425

TABLE IV

TELCO HEADLINE: PER-KPI MACRO METRICS FOR THE BEST SELF-TUNED SINGLETON, THE CANONICAL SHAP-σ PIPELINE AT THE PEAK RASHOMON SIZE, AND THE TWO RELEASED DEEP BASELINES. BOLD = BEST PER COLUMN.

Method	F_1	P	R
DC-VAE [2]	0.134	—	—
C-MTAD-GAT [3]	0.160	—	—
self-tuned LOF (best singleton)	0.419	0.460	**0.452**
SHAP-σ (K_R=4, Mixed)	**0.464**	**0.605**	0.425

V. CONCLUSIONS

We presented SHAP-σ, a label-free pipeline for multivariate cellular network anomaly detection that promotes the cross-detector standard deviation of SHAP attributions from a post-hoc explanation into a first-class attribution-evidence signal. The methodological contributions are: (i) label-free self-tuning of a K=10 unsupervised detector pool (including an AutoEncoder and OCSVM alongside the eight standard PyOD detectors); (ii) a closed-form Rashomon pair score $\frac{q_i+q_j}{2}(1-\rho_{ij})$ combining a Mass–Volume proxy with a cardinality veto; and (iii) a per-KPI σ gate that filters anchor flags by the cross-detector spread of TreeSHAP attributions. On the TELCO cellular benchmark a K_R sweep peaks at K_R=4 with the self-tuned quartet {ECOD, LOF, GMM, OCSVM}, reaching per-KPI macro F_1=0.464 (precision 0.605, recall 0.425), +0.045 over the best self-tuned singleton. The pipeline exceeds the released deep baselines DC-VAE and C-MTAD-GAT [2], [3] by 2.9–3.5× on the same evaluation window, while remaining transparent and reproducible per-KPI.

a) Limitations and future work.: Eq. (5) is recall-monotone with respect to the anchor singleton; the disjunctive feature-wise gate is most informative on per-feature labels (as in TELCO) and degenerates under OR-aggregation for point-level multivariate evaluation, which we defer to a journal extension on extended cellular benchmarks. We also plan to learn $(K_R, \mathcal{S})$ label-free from synthetic anomaly injection [8], [10], to compare against recent deep label-free baselines such as MTGFlow [33], and to extend σ-SHAP to cross-window variance for streaming deployment.

REFERENCES

[1] N. Goix, "How to evaluate the quality of unsupervised anomaly detection algorithms?" in *ICML 2016 Workshop on Anomaly Detection*, 2016.

[2] G. García González, P. Casas, A. Fernández, and G. Gómez, "One model to find them all: A unified anomaly detection framework for multivariate time series in mobile networks," *IEEE Transactions on Network and Service Management*, 2023.

[3] S. Malacarne, E. Hoel-Høiseth, E. Aune, D. Z. Biro, and M. Ruocco, "Context-aware graph attention for unsupervised telco anomaly detection," 2026, arXiv:2604.27172v1.

[4] M. Q. Ma, Y. Zhao, X. Zhang, and L. Akoglu, "The need for unsupervised outlier model selection: A review and evaluation of internal model selection strategies," *ACM SIGKDD Explorations Newsletter*, vol. 25, no. 1, pp. 19–35, 2023.

[5] L. Breiman, "Statistical modeling: The two cultures," *Statistical Science*, vol. 16, no. 3, pp. 199–231, 2001.

[6] L. Semenova, C. Rudin, and R. Parr, "On the existence of simpler machine learning models," in *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2022, pp. 1827–1858.

[7] A. Fisher, C. Rudin, and F. Dominici, "All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously," *Journal of Machine Learning Research*, vol. 20, no. 177, pp. 1–81, 2019.

[8] Y. Zhao, R. A. Rossi, and L. Akoglu, "Automatic unsupervised outlier model selection," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[9] Y. Zhao and L. Akoglu, "Toward unsupervised outlier model selection," in *Proceedings of the 22nd IEEE International Conference on Data Mining (ICDM)*, 2022.

[10] M. Goswami, C. Challu, L. Callot, L. Minorics, and A. Kan, "Unsupervised model selection for time-series anomaly detection," in *International Conference on Learning Representations (ICLR)*, 2023.

[11] Y. Zhao, Z. Nasrullah, and Z. Li, "PyOD: A python toolbox for scalable outlier detection," *Journal of Machine Learning Research*, vol. 20, no. 96, pp. 1–7, 2019.

[12] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[13] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

[14] A. Levy, P. Saves, M. Garouani *et al.*, "Analyzing SHAP attributions for anomaly-detection ensembles," in *Advances in Intelligent Data Analysis (IDA)*, ser. LNCS. Springer, 2026.

[15] J. Audibert, P. Michiardi, F. Guyard, S. Marti, and M. A. Zuluaga, "USAD: UnSupervised anomaly detection on multivariate time series," in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*, 2020, pp. 3395–3404.

[16] S. Tuli, G. Casale, and N. R. Jennings, "TranAD: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[17] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[18] Y. Yang, C. Zhang, T. Zhou, Q. Wen, and L. Sun, "DCdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023, pp. 3033–3045.

[19] C. C. Aggarwal and S. Sathe, *Outlier Ensembles: An Introduction*. Springer, 2017.

[20] A. Zimek, R. J. G. B. Campello, and J. Sander, "Ensembles for unsupervised outlier detection: Challenges and research questions," in *ACM SIGKDD Explorations Newsletter*, vol. 15, no. 1, 2014, pp. 11–22.

[21] L. I. Kuncheva and C. J. Whitaker, "Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy," *Machine Learning*, vol. 51, no. 2, pp. 181–207, 2003.

[22] Z. Li, Y. Zhao, X. Hu, N. Botta, C. Ionescu, and G. H. Chen, "ECOD: Unsupervised outlier detection using empirical cumulative distribution functions," *IEEE Transactions on Knowledge and Data Engineering*, 2022.

[23] Z. Li, Y. Zhao, N. Botta, C. Ionescu, and X. Hu, "COPOD: Copula-based outlier detection," in *2020 IEEE International Conference on Data Mining (ICDM)*, 2020, pp. 1118–1123.

[24] M. Goldstein and A. Dengel, "Histogram-based outlier score (HBOS): A fast unsupervised anomaly detection algorithm," in *KI-2012: Poster and Demo Track*, 2012, pp. 59–63.

[25] S. Ramaswamy, R. Rastogi, and K. Shim, "Efficient algorithms for mining outliers from large data sets," in *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, 2000, pp. 427–438.

[26] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: identifying density-based local outliers," in *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, 2000, pp. 93–104.

[27] M.-L. Shyu, S.-C. Chen, K. Sarinnapakorn, and L. Chang, "A novel anomaly detection scheme based on principal component classifier," in *Proceedings of the IEEE Foundations and New Directions of Data Mining Workshop (ICDM)*, 2003, pp. 172–179.

[28] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *2008 Eighth IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422.

[29] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Computation*, vol. 13, no. 7, pp. 1443–1471, 2001.

[30] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 1996, pp. 226–231.

[31] E. Black, K. Leino, and M. Fredrikson, "Selective ensembles for consistent predictions," in *International Conference on Learning Representations (ICLR)*, 2022.

[32] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[33] Q. Zhou, S. He, H. Liu, J. Chen, and W. Meng, "Label-free multivariate time series anomaly detection," *arXiv preprint arXiv:2312.11549*, 2024.